

# Lab 6 Prelab

## Reflection on Assessing and Expanding Background Knowledge Sections

The assessing and expanding background knowledge sections helped clarify how coordinate frames, translations, and rotations are used to describe relationships between different parts of a robot. Rather than defining everything in a single global frame, positions and orientations are described relative to parent frames, which makes it easier to model complex systems. Understanding how frames shift and rotate using the right-hand rule was especially important, since it directly affected how joints and links behaved in the URDF. Applying these concepts during the URDF creation showed how incorrect frame assumptions can lead to unexpected motion or misplaced geometry, reinforcing why a solid understanding of coordinate frames is essential when building and debugging robotic models.

## One Material Created in Activity

**Here is a screenshot of the material created in the activity.**

```
<!-- MATERIAL -->
<material name="wood_brown">
  <color rgba="0.6 0.3 0.15 1.0"/>
</material>
```

I added a brown material to the box to visually distinguish it and satisfy the modification requirement.

## URDF File and Screenshot in Rviz (With Joint State Publisher Visible) of the Completed Box With Modification

**Here is my URDF file.**

```
<?xml version="1.0"?>
<robot name="box">

  <!-- MATERIAL -->
  <material name="wood_brown">
    <color rgba="0.6 0.3 0.15 1.0"/>
  </material>

  <!-- BASE -->
  <link name="base_link">
    <visual>
      <geometry>
        <box size="1 1 0.05"/>
      </geometry>
```

```

    <material name="wood_brown"/>
    <origin xyz="0 0 0" rpy="0 0 0"/>
  </visual>
</link>

<!-- RIGHT -->
<link name="right_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <material name="wood_brown"/>
    <origin xyz="0 0 0.5" rpy="1.5708 0 0"/>
  </visual>
</link>

<joint name="right_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="right_link"/>
  <origin xyz="0 0.5 0" rpy="0 0 0"/>
</joint>

<!-- LEFT -->
<link name="left_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <material name="wood_brown"/>
    <origin xyz="0 0 0.5" rpy="1.5708 0 0"/>
  </visual>
</link>

<joint name="left_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="left_link"/>
  <origin xyz="0 -0.5 0" rpy="0 0 0"/>
</joint>

<!-- BACK -->
<link name="back_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <material name="wood_brown"/>
    <origin xyz="0 0 0.5" rpy="1.5708 0 1.5708"/>
  </visual>
</link>

<joint name="back_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="back_link"/>
  <origin xyz="-0.5 0 0" rpy="0 0 0"/>
</joint>

<!-- FRONT -->
<link name="front_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>

```

```

</geometry>
<material name="wood_brown"/>
<origin xyz="0 0 0.5" rpy="1.5708 0 1.5708"/>
</visual>
</link>

<joint name="front_to_bottom" type="fixed">
  <parent link="base_link"/>
  <child link="front_link"/>
  <origin xyz="0.5 0 0" rpy="0 0 0"/>
</joint>

<!-- TOP (LID) -->
<link name="top_link">
  <visual>
    <geometry>
      <box size="1 1 0.05"/>
    </geometry>
    <material name="wood_brown"/>
    <origin xyz="0.5 0 0" rpy="0 0 0"/>
  </visual>
</link>

<joint name="top_to_back" type="revolute">
  <axis xyz="0 -1 0"/>
  <parent link="back_link"/>
  <child link="top_link"/>
  <origin xyz="0 0 1" rpy="0 0 0"/>
  <limit effort="1000" lower="0" upper="1.57" velocity="0.5"/>
</joint>

</robot>

```

**Here is a screenshot of the Completed URDF box visualized in RViz with joint state publisher enabled.**

